

(a) Fixed ECM: all training states, one weight vector

$$\begin{array}{ccc}
 \mathbf{M}_{\text{int}} = \begin{bmatrix} (\mathbf{c}_1^1)^T & \cdots & (\mathbf{c}_1^{N_c})^T \\ \vdots & & \vdots \\ (\mathbf{c}_{N_{\text{el}}}^1)^T & \cdots & (\mathbf{c}_{N_{\text{el}}}^{N_c})^T \end{bmatrix} & \xrightarrow[\text{add constraint}]{\begin{array}{c} \text{Truncated SVD} \\ \text{transpose} \end{array}} & \mathbf{G}_{\text{cub}} = \begin{bmatrix} \mathbf{U}_{\text{cub}}^T \\ \mathbf{c}_{\perp}^T \end{bmatrix} \\
 \begin{array}{c} \text{Rows: elements} \\ \text{Columns: state-component pairs} \\ N_{\text{el}} \times (dN_c) \end{array} & & \begin{array}{c} \text{Columns: candidate elements} \\ \text{Rows: modes} \\ + \text{orthogonalized constant} \\ (r_c + 1) \times N_{\text{el}} \end{array}
 \end{array}
 \xrightarrow{\text{Greedy selection}}
 \begin{array}{c}
 \mathbf{G}_{\text{cub}} \mathbf{w} \simeq \mathbf{b}_{\text{cub}}, \quad \mathbf{w} \geq 0 \\
 \mathbf{b}_{\text{cub}} = \mathbf{G}_{\text{cub}} \mathbf{1} \\
 \text{One } \mathbf{w} \text{ for all retained modes} \\
 w_e = 0 \text{ outside the support}
 \end{array}$$

(b) MAW-ECM weight-field fit: one system per state, common support

$$\begin{array}{ccc}
 \text{Columns: selected elements } e_1, \dots, e_m & & \\
 \begin{bmatrix} \mathbf{c}_{e_1}^s & \cdots & \mathbf{c}_{e_m}^s \\ 1 & \cdots & 1 \end{bmatrix} & \underbrace{\begin{bmatrix} w_{e_1}(\mathbf{q}^s) \\ \vdots \\ w_{e_m}(\mathbf{q}^s) \end{bmatrix}}_{\mathbf{w}(\mathbf{q}^s) \text{ State-dependent weights}} & \simeq \begin{bmatrix} \sum_{e \in \mathcal{E}} \mathbf{c}_e^s \\ N_{\text{el}} \end{bmatrix} \\
 \mathbf{A}^s: d \text{ component rows} + \text{normalization} & & \mathbf{b}^s: \text{full-mesh sum} \\
 (d+1) \times m & & (d+1) \times 1
 \end{array}$$

$$\boxed{\mathbf{A}^1 \mathbf{w}(\mathbf{q}^1) \simeq \mathbf{b}^1} \quad \dots \quad \boxed{\mathbf{A}^{N_c} \mathbf{w}(\mathbf{q}^{N_c}) \simeq \mathbf{b}^{N_c}}$$

After pruning: one weight ANN shared across all training states

$$\mathbf{q} \xrightarrow{N_r} \ell_r \xrightarrow{N_{\text{el}} \text{ softmax}} \mathbf{w}(\mathbf{q})$$

(c) Adaptive stress example of (b): four first Piola components + normalization

$$\begin{array}{ccc}
 \mathbf{p}_e^s = \frac{1}{|Y|} \sum_{g \in e} \omega_g \mathbf{P}_{\mu, g}^s, & \bar{\mathbf{P}}^s = \sum_{e \in \mathcal{E}} \mathbf{p}_e^s, & \mathcal{Z}_s = \{e_1, \dots, e_{m_s}\} \\
 \begin{bmatrix} p_{e_1, 11}^s & \cdots & p_{e_{m_s}, 11}^s \\ p_{e_1, 12}^s & \cdots & p_{e_{m_s}, 12}^s \\ p_{e_1, 21}^s & \cdots & p_{e_{m_s}, 21}^s \\ p_{e_1, 22}^s & \cdots & p_{e_{m_s}, 22}^s \\ 1 & \cdots & 1 \end{bmatrix} & \mathbf{v}(\mathbf{q}^s) \simeq & \begin{bmatrix} \bar{P}_{11}^s \\ \bar{P}_{12}^s \\ \bar{P}_{21}^s \\ \bar{P}_{22}^s \\ N_{\text{el}} \end{bmatrix} \\
 \mathbf{A}_{\sigma}^s \in \mathbb{R}^{5 \times m_s}: \text{selected contributions} & & \mathbf{b}_{\sigma}^s \in \mathbb{R}^5: \text{full-mesh target} \\
 & & N_{\text{el}} \text{ softmax}(\mathcal{N}_s(\mathbf{q}^s))
 \end{array}$$

Fixed stress ECM: stack these contributions in $\mathbf{M}_{\text{int}} \in \mathbb{R}^{N_{\text{el}} \times 4N_c}$, then follow panel (a).